

1.Question:

X and Y are developing a new product idea and must decide whether to build a physical prototype or use simulation software to test their concept. With a tight budget and a limited timeline, they're unsure which approach will be more efficient. Justify the needs of simulation in this case **[4 Marks]**

Answer:

Since X and Y have a limited budget and time, using simulation software is the better option than building a physical prototype.

Justification:

1. **Low Cost:** Simulation avoids the expense of making a physical model.
2. **Saves Time:** Different designs can be tested quickly.
3. **Easy Modification:** Changes can be made without rebuilding.
4. **Risk-Free Testing:** Problems can be identified before manufacturing.
5. **Better Decision Making:** Simulation helps compare different designs and choose the best one.

Conclusion: Simulation is the most efficient approach because it is **faster, cheaper, and safer.**

2(a) Question:

Classify levels of simulation with proper explanation. **[3 Marks]**

Answer:

There are **five levels of simulation:**

1. Block-Level Simulation

- Tests **individual functional blocks** of a circuit.
- Helps find errors easily.
- Blocks can be reused in other designs.

3. Memory Simulation

- Tests **memory devices** such as **SRAM, DRAM, and NAND Flash.**
- Checks memory timing and performance.

2. Chip-Level Simulation

- Tests the **entire IC (chip)** as one unit.
- Ensures the chip works correctly before fabrication.

4. Mixed-Signal Simulation

- Combines **analog and digital simulation.**
- Used for **SoC (System-on-Chip)** and embedded systems.

5. RF (Radio Frequency) Simulation

- Tests **RF circuits** and communication systems.
- Used for components like **mixers, amplifiers, and PLLs**.

2(b) Question:

Summarize the types of circuit simulation. **[3 Marks]**

Answer:

Circuit Simulation is mainly divided into **three types**:

1. Analog Circuit Simulation

- Used for **analog circuits and continuous signals**.
- Analyzes **DC, AC, and Transient** behavior.
- Provides highly accurate results.
- Popular simulators: **SPICE** and **FastSPICE**.

2. Digital Circuit Simulation

- Used for **digital circuits**.
- Works with **two logic levels: Logic 0 and Logic 1**.
- Uses **HDL (Hardware Description Language)** such as **Verilog** and **VHDL**.

3. Mixed-Mode Circuit Simulation

- Combination of **Analog and Digital Simulation**.
- Used for circuits containing both analog and digital parts.
- Helps in designing and analyzing mixed electronic systems.

Extra question:

Popular Simulation Software

- **SPICE**
- **ngspice**
- **LTspice (SPICE)**
- **PSIM**
- **Multisim (SPICE)**
- **Cadence Spectre**
- **Synopsys PrimeSim (SPICE and FastSPICE)**

Falstad Circuit Simulator Algorithm (5 Marks)

1. Initialize

- Set the number of nodes, voltages, currents, and time step.

2. Update Dynamic Element

- Convert capacitors and inductors into equivalent circuits.

3. Linearize Nonlinear Devices

- Simplify diodes and transistors for easy calculation.

4. Assemble Circuit Equation Matrix and Solve

- Form the circuit equations and calculate node voltages and currents.

5. Update Time-Dependent States

- Update capacitor voltages and inductor currents for the next time step.

6. Render Visualization

- Display node voltages, branch currents, and waveforms.

7. Advance Time

- Repeat the above steps until the simulation ends.



Falstad Circuit Simulator Algorithm

1. **Initialize:** Number nodes, initialize voltages, capacitor voltages, inductor currents, and timestep.
2. **Update dynamic element :** Convert capacitors and inductors into equivalent resistances/current sources for this time step.
3. **Linearize nonlinear devices:** Apply Newton–Raphson iteration for diodes, transistors, etc.
4. **Assemble Circuit Equation matrix and solve :** Include resistors, voltage/current sources, dynamic equivalents, and linearized nonlinear elements and Compute node voltages.
5. **Update time-dependent states:** Capacitor voltages and inductor currents for the next timestep.
6. **Render visualization:** Node voltages (color), branch currents (animated dots), waveforms.
7. **Advance time:** Repeat steps 2–8 until simulation end.

Memory Trick

$$I \rightarrow U \rightarrow L \rightarrow A \rightarrow U \rightarrow R \rightarrow A$$

Question: When Simulation is Not Appropriate (5 Marks)

Simulation is **not appropriate** in the following situations:

1. When the problem can be solved using common sense

If the solution is simple and obvious, simulation is unnecessary.

2. When the problem can be solved analytically

If mathematical formulas or calculations can solve the problem, simulation is not needed.

3. When direct experiments are easier

If the real system can be tested easily and safely, direct testing is a better choice.

4. When simulation cost is higher than the benefit

If the cost of simulation is greater than the expected savings, it should not be used.

5. When time or resources are not available

If there is not enough time, software, or skilled people, simulation may not be practical.

6. When the system is too complex or undefined

If the system behavior cannot be clearly defined or modeled, simulation results may be unreliable.

7. When the model cannot be verified and validated

If the simulation model cannot be tested for accuracy, the results cannot be trusted.

Question : Advantages and Disadvantages of Simulation (4 Marks)

Advantages of Simulation

1. **Saves Time** – Tests the circuit before building it.
2. **Reduces Cost** – Avoids expensive hardware mistakes.
3. **Finds Errors Early** – Detects problems before fabrication.
4. **Easy to Modify** – Circuit design can be changed and tested quickly.

Disadvantages of Simulation

1. **Results May Not Be 100% Accurate** – Depends on the simulation model.
2. **Requires Skilled Users** – Needs knowledge of simulation software.
3. **Can Be Time-Consuming** – Complex circuits take longer to simulate.
4. **Software Can Be Expensive** – Some simulation tools require costly licenses.

3(a)

Suppose you want to propose a new system for solving a problem using simulation.

Summarize the various steps for this simulation study with a proper flowchart. [5 Marks]

Answer:

Steps of Simulation Study

1. Identify the problem.
2. Define objectives.
3. Collect data.
4. Develop the simulation model.
5. Verify and validate the model.
6. Run the simulation.
7. Analyze the results.
8. Make the final decision.

Flowchart

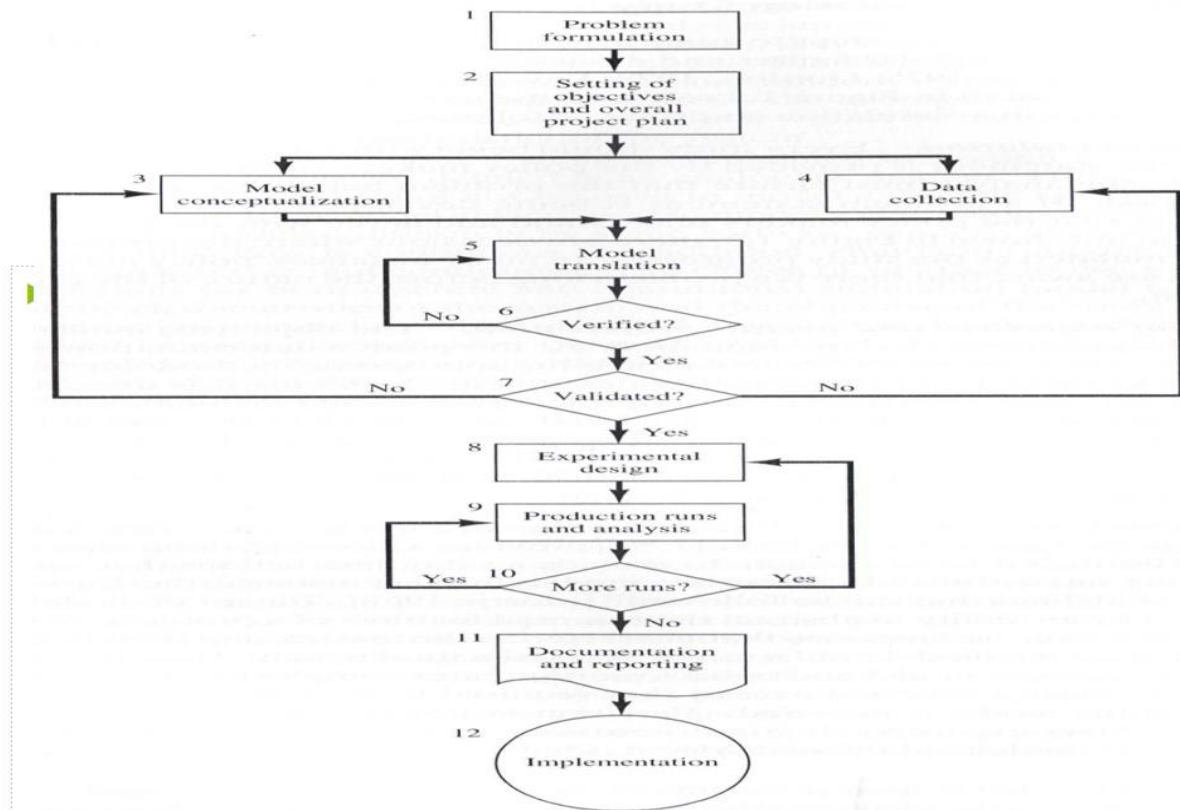


Figure 1.3. Steps in a simulation study. 9

Question 3(b) [2 Marks]

Explain the components of your proposed system with example.

Answer:

Components

arrive, wait for service, and leave after completing transaction

1. **Input** – Data given to the system.
2. **Simulation Model** – Represents the real system.
3. **Processor** – Performs calculations.
4. **Output** – Displays simulation results.

Example:

ATM queue simulation where customers

Components of a System

A system has **5 main components**:

1. **Entity** – The object or person in the system.
2. **Attribute** – The property or characteristic of an entity.
3. **Activity** – The action performed in the system.
4. **Event** – Something that changes the system.
5. **State Variable** – Shows the current condition of the system.

System	Entity	Attribute	Activity	Event	State Variable
Banking	Customers	Account balance	Deposit money	Arrival / Departure	Number of busy tellers, customers waiting
Rapid Rail	Riders	Origin, Destination	Traveling	Arrival at station	Riders waiting, riders in transit
Production	Machines	Speed, Capacity	Welding, Stamping	Breakdown	Machine status (Busy/Idle/Down)
Communication	Messages	Length, Destination	Transmitting	Arrival at destination	Messages waiting to send
Inventory	Warehouse	Capacity	Withdrawing items	Demand	Inventory level, backlogged demand

System Categorization (2–3 Marks)

Systems are of **two types**:

1. Discrete System

- Changes only at **specific points in time**.
- State changes in **steps**.
- **Example:** Number of customers in a bank.

► Systems can be categorized as discrete or continuous.

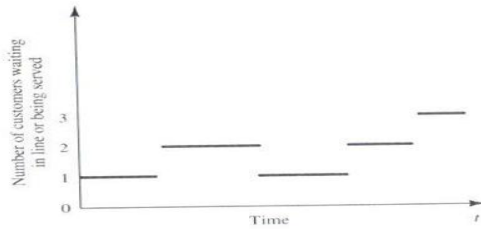


Figure 1.1. Discrete-system state variable.

2. Continuous System

- Changes **continuously with time**.
- State changes **smoothly**.
- **Example:** Water level in a dam.

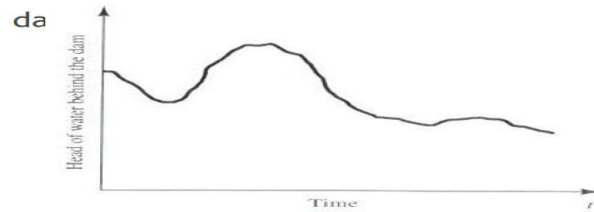


Figure 1.2. Continuous-system state variable.

Steps in a Simulation Study (Short Answer – 3/4 Marks)

1. **Problem Formulation:** Define the problem clearly.
2. **Setting Objectives:** Set the goals and prepare the project plan.
3. **Model Conceptualization:** Develop a simple model of the real system.
4. **Data Collection:** Collect the required input data.
5. **Model Translation:** Convert the model into a simulation program.
6. **Verification:** Check whether the program works correctly.
7. **Validation:** Ensure the model represents the real system accurately.
8. **Experimental Design:** Decide the simulation conditions and number of runs.
9. **Production Runs and Analysis:** Run the simulation and analyze the results.
10. **More Runs:** Perform additional runs if necessary.
11. **Documentation and Reporting:** Prepare documentation and report the results.
12. **Implementation:** Apply the simulation results to the real system.

Monte Carlo Simulation Steps (2 Marks)

To understand this Technique, this is break down into 5 Steps.

1. Establishing Probability Distribution. (Probability)
2. Cumulative Probability Distribution.
3. Setting Random Number Intervals.
4. Generating Random Numbers.
5. To Find the answer of question asked using the above four Steps.

Solution:-

Table - I

0 - 39
40 - 54

Type	Probability	C. Probability	Random No.
Filling	0.40	0.40	00 - 39
Crown	0.15	0.55	40 - 54
Cleaning	0.15	0.70	55 - 69
Extracting	0.10	0.80	70 - 79
Checkup	0.20	1.00	80 - 99

Table - II

Patient	Scheduled Arrival	Random Number	Category	Service Time Needed
1.	8:00	40	Crown	60 min
2.	8:30	82	Check-up	15 min
3.	9:00	11	Filling	45 min
4.	9:30	34	Filling	45 min
5.	10:00	25	Filling	45 min
6.	10:30	66	Cleaning	15 min
7.	11:00	17	Filling	45 min
8.	11:30	79	Extracting	45 min

Patient	Arrival	Service Start	Service Duration	Service Ends	Waiting (in Minutes)	Idle time
1	8:00	8:00	60 min	9:00	0	0
2	8:30	9:00	15 min	9:15	30	0
3	9:00	9:15	45 min	10:00	15	0
4	9:30	10:00	45 min	10:45	30	0
5	10:00	10:45	45 min	11:30	45	0
6	10:30	11:30	15 min	11:45	60	0
7	11:00	11:45	45 min	12:30	45	0
8	11:30	12:30	45 min	1:15	60	0

Simulation is the process of creating a **model of a real-world system** to study, test, and predict its behavior **without using the actual system**.

A grocery store owner wants to calculate the total price of 20 items purchased by a customer. One employee suggests using simulation software, while another suggests using a calculator. With limited time, which approach is more appropriate? Justify your answer.

Answer:

Using a **calculator** is more appropriate because **simulation is not appropriate** in this case.

Justification:

1. The problem is simple.
2. The result can be calculated directly.
3. Simulation wastes time.
4. No real-world system needs to be modeled.
5. A calculator gives the exact answer quickly.

Conclusion: Simulation should not be used for simple calculations.

Scenario-Based Question 2 (4 Marks)

A teacher wants to calculate the average marks of 50 students. One student suggests developing a simulation model, while another recommends using Excel. Which method is more suitable? Justify your answer.

Answer:

Using **Excel** is more suitable because simulation is unnecessary.

Justification:

1. Average can be calculated directly.
2. No uncertainty exists.
3. Simulation increases complexity.
4. Excel provides accurate results quickly.
5. It saves both time and effort.

Conclusion: Simulation is not appropriate.

Scenario-Based Question 3 (4 Marks)

A bank wants to calculate the monthly interest of customers using a fixed formula. Some employees propose using simulation software. Do you agree? Justify your answer.

Answer:

No, I do not agree.

Justification:

1. Interest is calculated using a fixed formula.
2. Exact results are available directly.
3. Simulation is unnecessary.
4. It wastes time and resources.
5. Banking software can perform the calculation easily.

Conclusion: Simulation is not appropriate for fixed mathematical calculations.

Scenario-Based Question 4 (4 Marks)

A college office wants to prepare students' GPA using their marks. One staff member recommends building a simulation model, while another recommends using spreadsheet software. Which method is better? Explain your answer.

Answer:

Spreadsheet software is the better choice.

Justification:

1. GPA calculation follows a fixed formula.
2. No simulation model is required.
3. Results can be obtained instantly.
4. Simulation increases unnecessary work.
5. It is more efficient to use Excel.

Conclusion: Simulation should not be used.

Scenario-Based Question 5 (4 Marks)

A factory manager wants to know the total production of machines in one day by adding the production values. One engineer suggests using simulation software. Is this the correct approach? Justify your answer.

Answer:

No, it is not the correct approach.

Justification:

1. Total production can be calculated by simple addition.
2. No complex system needs analysis.
3. Simulation wastes time.
4. Direct calculation is easier.
5. The result is exact without simulation.

Conclusion: Simulation is not appropriate.